Parent Guide — middle / module-11-multiplication-near-base

IKS Curriculum

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Parent Guide — Module 11: Multiplication Near a Base (Nikhilam Method)

Your child's age: Middle (Grades 6–8) · **Module length:** 10 sessions × 45 min

What your child is learning

A mental-multiplication shortcut for two numbers that are both close to a power of 10 (10, 100, 1000). The Sanskrit name is *Nikhilam* — short for *Nikhilam Navataścaramaṁ Daśataḥ*, "all from 9 and the last from 10." Same sūtra that powered Module 9's subtraction.

The trick in 30 seconds:

> To multiply 97 × 96 (both near 100): 1. Find the **deficits**: 97 is 3 below 100 (−3); 96 is 4 below 100 (−4). 2. **Cross-subtract** for the "left" part: 97 − 4 = 93 (or equivalently 96 − 3 = 93). 3. **Multiply the deficits** for the "right" part: 3 × 4 = 12. 4. Stick them together: **9312**.
>
> Check on a calculator if you like. It works because (100 − 3)(100 − 4) = 100 × 93 + 12.

Baseline → retest: On Day 1, your child times themselves doing `97 × 96` by long multiplication. On Day 9, they retest the same problem under the Nikhilam method. The visible speedup (often 4–10×) is the module's motivation.

When does this method help?

- When **both** numbers are within ~5–10 of the same power of 10 (so: near 10, near 100, near 1000).
- **Not** when one of the numbers is "random" (e.g. 47 × 53). The honest framing on Day 8 is: *standard long multiplication is the better tool most of the time.*

A note on "Vedic mathematics"

Important honest framing: - The *Nikhilam* sūtra and the multiplication technique are published in Bharati Krishna Tirthaji's *Vedic Mathematics* (1965, posthumous). The math is real and useful. - Tirthaji claimed the sūtras came from an *Atharva-veda Pariśiṣṭa* (Vedic appendix-text). **No other scholar has produced this manuscript.** Mainstream historians (Dani 1993, Plofker 2009) consider the sūtras a 20th-century synthesis rather than ancient Vedic mathematics. - The cultural achievement is genuine. The history is just more recent than the name suggests.

Your child explores this honest framing on Day 7. At Senior band (Grade 9–12) they'll read the scholarly debate and form their own informed position.

What to expect this fortnight

Daily warm-up: 1 minute of "complement to 10" fluency (teacher says "7," class says "3"); from Day 4 a "complement to 100" round is added.

Speed drills: paired, self-timed. No public leaderboards.

Project (Days 9–10): Your child chooses one of: - A 60-second teaching video (Grade 4 audience). - A printed 20-problem workbook with worked solutions. - A comparative speed-test data project (5–10 family/friends timed on long multiplication vs. Nikhilam).

How to help

Helpful: - For the speed-test project: be a willing test subject. Let your child time you doing `97 × 96` the long way, then time them doing it the Nikhilam way. - 5-minute home drill: call out 2-digit numbers near 100 (97, 96, 94, 88, 102, 104, 99 × 98, etc.) and let your child answer the deficits or excesses, then the products. - Try the method yourself on `103 × 104`. You'll get **10712** in under 10 seconds. Once you've felt it, you'll appreciate what your child is learning.

Not helpful: - Insisting the standard algorithm is the "right" way. Both methods give the same answer. - Doing their project. Even the workbook has to be in their handwriting / their drawings. - Hyping the "ancient Vedic" framing. The math is real; the antiquity is contested.

What success looks like

By end of module, your child can: - Multiply any 2-digit pair near 100 mentally in under 5 seconds (e.g. 96 × 97, 103 × 104, 102 × 98). - Handle the right-part overflow (88 × 89 → carry needed). - Handle the mixed-sign case (102 × 98). - Multiply 3-digit pairs near 1000 (998 × 997) with paper. - Explain *why* the method works using one line of algebra. - Tell you honestly when long multiplication is the better tool.

Citation standard

Projects must cite the source PDF (e.g. *Vedic Mathematics Batch1.pdf*) and may also name the sūtra (*Nikhilam Navataścaramaṁ Daśataḥ*). The full citation chain is documented in the module's `sources.md`.

Want to go deeper?

Senior band (Grade 9–12) generalises Nikhilam to non-decimal bases AND tackles the historicity debate in depth. Preview: `curriculum/senior/module-11-multiplication-near-base/README.md`.

10-Day Lesson Plan

Module 11 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	The Hook — 97 × 96 in three seconds	`days/day-01-hook.md`	Baseline timed test
2	The Setup — deficits from 10	`days/day-02-deficits-from-ten.md`	Pattern recall + 2-min sprint
3	Both Below 100	`days/day-03-both-below-100.md`	`quizzes/formative.md` Part A
4	Both Above 100	`days/day-04-both-above-100.md`	`quizzes/formative.md` Part B
5	Mixed Cases (one above, one below)	`days/day-05-mixed-cases.md`	`activities/activity-01-mixed-signs.md`
6	Near 1000 — padding the right part	`days/day-06-near-1000.md`	`quizzes/formative.md` Part C
7	Why It Works — the algebra	`days/day-07-why-it-works.md`	Reflection paragraph
8	When To Use It — vs. long multiplication	`days/day-08-when-to-use.md`	Decision quiz
9	Speed Sprints + Project Start	`days/day-09-speed-and-project.md`	`activities/activity-02-speed-sprint.md`
10	Final Assessment + Showcase	`days/day-10-final-assessment.md`	`quizzes/summative.md` + project

Pacing notes

- Day 1’s baseline (timing each student on `97 × 96` solved by long multiplication) is the motivational anchor. Day 9 retests the same problem set under the Nikhilam method; the visible speedup is the module’s payoff.
- Days 2–6 are the core. If the class catches on fast, you can combine Days 3+4 (both-below + both-above) into one period and gain a day for Day 9’s sprint.
- Day 7 (the algebra) is non-negotiable for Middle band — the *why* prevents students from treating the method as a magic trick.

- Day 8 (“when to use it”) balances enthusiasm with honesty: this method is only fast in a narrow regime. Students should walk away knowing when to switch back to long multiplication.

Daily warm-up (recurring)

Every class starts with **1 minute of “complement to 10” fluency**: teacher calls out a single digit; students call back the deficit (e.g. teacher says “7,” class chants “3”). This is the same warm-up as Module 5 / Module 9, reused — it builds the foundational reflex.

Daily warm-up (Day 4 onward)

Add a **1-minute “complement to 100”** round: teacher says “97,” class chants “3.” Teacher says “88,” class chants “12.” This builds the deficit-from-100 reflex that drives 2-digit Nikhilam.

What students leave the module with

- Mental Nikhilam multiplication of 2-digit numbers near 100 in under 5 seconds.
- A clear sense of when this method is faster than long multiplication, and when it is not.
- Algebraic understanding: they can prove the method with one line of algebra.
- Awareness that “Vedic mathematics” is a 20th-century pedagogical project, not strictly ancient — see `teacher-notes.md`.
- One artefact (their project) they can share with younger students or family.

Homework

Homework — Module 11 Nikhilam Multiplication (Middle)

Day 1 → Day 2

Task: Practice the deficit-from-10 reflex. Write the deficit of each from 10: 7 → __ ; 8 → __ ; 9 → __ ; 6 → __ ; 5 → __ ; 4 → __ ; 3 → __ ; 2 → __ ; 1 → __.

Do all 9 in under 10 seconds. Repeat 3 times.

Time: 5 min.

Day 2 → Day 3

Task: Solve 10 single-digit × single-digit problems (both 6–9) using Nikhilam. Self-check against your times table. Specifically: `7×8`, `8×9`, `9×9`, `6×7`, `6×9`, `7×7`, `8×8`, `6×8`, `9×7`, `9×8`.

Time: 10 min.

Day 3 → Day 4

Task: Solve 10 problems where both factors are between 85 and 99. Write deficits, left, padded right, and answer for each.

Time: 15 min.

Day 4 → Day 5

Task: Mixed sheet — 10 problems, half both-below-100 and half both-above-100. Decide the case before solving.

Time: 15 min.

Day 5 → Day 6

Task: Bring a list of any 5 multiplication problems from your daily life (price tags in a shop, distances, anything). For each, decide: Nikhilam-suitable or not? Justify in 1 sentence.

Time: 10 min.

Day 6 → Day 7

Task: Submit your project plan paragraph in writing (4–6 sentences). Choose format. Choose featured example.

Time: 15 min.

Day 7 → Day 8

Task: Write a 1-paragraph explanation suitable for a Grade 7 student of the algebraic identity `(B−a)(B−b) = B(B−a−b) + ab`. Include one worked example with the algebra shown.

Time: 15 min.

Day 8 → Day 9

Task: Practise your featured project example until you can present it without notes. Bring ALL project materials tomorrow.

Time: 10 min.

Day 9 → Day 10

Task: Final touches on your project. Make sure you have: the artefact, your script/notes, your 3-min presentation timed in your head, your citation visible on the artefact.

Time: 15 min.